User Scenarios List

Nr	Role	Name	Description
US_01	User	Search by title	User: Searches books by title so that they can quickly find a specific book.
US_02	User	Search by author	User: Searches books by author so that they can locate books from a specific writer.
US_03	User	Filter by category	User: Filters books by category so that they can see similar books together.
US_04	User	Filter by reading status	User: Filters books by reading status so that they can track their progress.
US_05	User	Sort by title	User: Sorts books by title so that they appear in alphabetical order.
US_06	User	Sort by author	User: Sorts books by author so that they are easier to browse.
US_07	User	Sort by publication year	User: Sorts books by publication year so that they can see the newest or oldest books.
US_08	User	Filter by rating	User: Filters books by rating so that they can see the best-rated books first.
US_09	User	Clear search filters	User: Clears all active search filters so that they can see the full book list again.
US_10	User	Search suggestions	User: Receives search suggestions while typing so that they can find books faster.

User Scenarios Extended

US_01: Search by title:

1. User navigates to the Catalog page.
2. User types a title or keyword in the search bar.
3. System queries the catalog for matching titles (case-insensitive).
4. System displays a list of matching books with title, author, and availability.
5. User selects a book to view full details.
6. If no results found: System displays 'No books found for your search'.
7. If search field is empty: System displays the full catalog.

US_02: Search by author:

1. User types an author name in the search bar.
2. System queries the catalog for books matching that author name.
3. System displays a list of books by that author.
4. User selects a book to view full details.
5. If no results found: System displays 'No books found for this author'.

US_03: Filter by category:

1. User opens the filter panel on the Catalog page.
2. User selects one or more categories from the list.
3. System filters the catalog to show only books in the selected categories.
4. User can combine category filter with other filters or search terms.
5. If no books match the selected category: System displays 'No books in this category'.

US_04: Filter by reading status:

1. User opens the filter panel.
2. User selects a reading status (e.g. Currently Borrowing, Returned, Reserved).
3. System filters the catalog based on the user's borrowing/reservation history.
4. User sees only books matching the selected status.
5. If no books match: System displays 'No books found for this status'.

US_05: Sort by title:

1. User opens the sort options on the Catalog page.
2. User selects 'Sort by Title'.
3. User chooses ascending (A–Z) or descending (Z–A) order.
4. System re-renders the catalog in the selected order.
5. Sort preference is maintained while browsing.

US_06: Sort by author:

1. User opens the sort options.
2. User selects 'Sort by Author'.
3. User chooses ascending or descending order.
4. System re-renders the catalog sorted by author name.
5. Sort preference is maintained while browsing.

US_07: Sort by publication year:

1. User opens the sort options.
2. User selects 'Sort by Publication Year'.
3. User chooses newest first or oldest first.
4. System re-renders the catalog sorted by publication year.
5. Sort preference is maintained while browsing.

US_08: Filter by rating:

1. User opens the filter panel.
2. User selects a minimum rating (e.g. 4 stars and above).
3. System filters the catalog to show only books with that rating or higher.
4. Books are displayed with their average rating.
5. If no books meet the rating threshold: System displays 'No books found for this rating'.

US_09: Clear search filters:

1. User has applied one or more filters or search terms.
2. User clicks the 'Clear Filters' button.
3. System resets all active filters and search terms.
4. System displays the full book catalog.
5. If no filters are active: Button is disabled or hidden.

US_10: Search suggestions:

1. User starts typing in the search bar.
2. System displays a dropdown list of matching book titles and authors in real-time.
3. User selects a suggestion from the dropdown.
4. System immediately filters the catalog based on the selected suggestion.
5. If no suggestions match: Dropdown is not shown.
6. If user ignores suggestions and presses Enter: Full search is performed.

Use Cases

UC_01: Search by title

UC Name	UC_01 — Search by title
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	The catalog contains at least one book.
Description of the Main Sequence	1. User navigates to the Catalog page. 2. User types a title or keyword in the search bar. 3. System performs a case-insensitive search on book titles. 4. System displays matching books with title, author, and availability. 5. User clicks a book to view full details.
Description of the Alternative Sequence	• No results: System displays 'No books found for your search'. • Empty search: System displays the full catalog.
Non functional requirements	Search results must appear within 1 second. Search must be case-insensitive.
Postconditions	User sees a list of books matching the entered title or keyword.

UC_02: Search by author

UC Name	UC_02 — Search by author
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	The catalog contains at least one book with an author assigned.
Description of the Main Sequence	1. User types an author name in the search bar. 2. System performs a case-insensitive search on author names. 3. System displays books by matching authors. 4. User selects a book to view full details.
Description of the Alternative Sequence	• No results: System displays 'No books found for this author'.
Non functional requirements	Search must return results within 1 second. Partial name matches must be supported.
Postconditions	User sees all books by the searched author.

UC_03: Filter by category

UC Name	UC_03 — Filter by category
Dependency	None

Actors	Primary actor: User

	Secondary actor: None
Preconditions	At least one category exists in the system.
Description of the Main Sequence	1. User opens the filter panel. 2. User selects one or more categories. 3. System filters the catalog to show only books in selected categories. 4. User can combine with other filters or search terms.
Description of the Alternative Sequence	• No books in category: System displays 'No books in this category'.
Non functional requirements	Filter must apply within 0.5 seconds. Multiple categories can be selected simultaneously.
Postconditions	Catalog shows only books belonging to the selected categories.

UC_04: Filter by reading status

UC Name	UC_04 — Filter by reading status
Dependency	User must be logged in
Actors	Primary actor: User Secondary actor: None
Preconditions	User is logged in and has at least one borrowing or reservation record.
Description of the Main Sequence	1. User opens the filter panel. 2. User selects a reading status (Currently Borrowing, Returned, Reserved). 3. System filters based on the user's personal loan and reservation history. 4. User sees only books matching the selected status.
Description of the Alternative Sequence	• No books match: System displays 'No books found for this status'.
Non functional requirements	Filter must apply within 1 second. Status options must reflect real-time data.
Postconditions	Catalog shows only books matching the user's selected reading status.

UC_05: Sort by title

UC Name	UC_05 — Sort by title
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	The catalog contains at least one book.

Description of the Main Sequence	1. User opens sort options. 2. User selects 'Sort by Title'. 3. User chooses ascending (A–Z) or descending (Z–A). 4. System re-renders the catalog in the selected order.

Description of the Alternative Sequence	• No preference selected: Default sort (A–Z) is applied.
Non functional requirements	Sort must apply instantly (< 0.5s). Sort preference is maintained while browsing.
Postconditions	Books are displayed in alphabetical order by title.

UC_06: Sort by author

UC Name	UC_06 — Sort by author
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	The catalog contains books with authors assigned.
Description of the Main Sequence	1. User opens sort options. 2. User selects 'Sort by Author'. 3. User chooses ascending or descending order. 4. System re-renders the catalog sorted by author name.
Description of the Alternative Sequence	• Books without author: Displayed at the end of the sorted list.
Non functional requirements	Sort must apply instantly (< 0.5s).
Postconditions	Books are displayed sorted by author name.

UC_07: Sort by publication year

UC Name	UC_07 — Sort by publication year
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	The catalog contains books with publication years.
Description of the Main Sequence	1. User opens sort options. 2. User selects 'Sort by Publication Year'. 3. User chooses newest first or oldest first. 4. System re-renders the catalog sorted by year.

Description of the Alternative Sequence	• Books without publication year: Displayed at the end of the list.
Non functional requirements	Sort must apply instantly (< 0.5s).
Postconditions	Books are displayed sorted by publication year.

UC_08: Filter by rating

UC Name	UC_08 — Filter by rating
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	At least one book has an average rating in the system.
Description of the Main Sequence	1. User opens the filter panel. 2. User selects a minimum rating (e.g. 4 stars and above). 3. System filters the catalog to show only books meeting or exceeding the threshold. 4. Books are displayed with their average rating.
Description of the Alternative Sequence	• No books meet threshold: System displays 'No books found for this rating'.
Non functional requirements	Filter must apply within 0.5 seconds.
Postconditions	Catalog shows only books with a rating equal to or above the selected threshold.

UC_09: Clear search filters

UC Name	UC_09 — Clear search filters
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	At least one filter or search term is currently active.
Description of the Main Sequence	1. User clicks the 'Clear Filters' button. 2. System resets all active filters and search terms. 3. System displays the full book catalog.
Description of the Alternative Sequence	• No active filters: Button is disabled or hidden.
Non functional requirements	Reset must be instant (< 0.3s).
Postconditions	All filters are cleared and the full catalog is displayed.

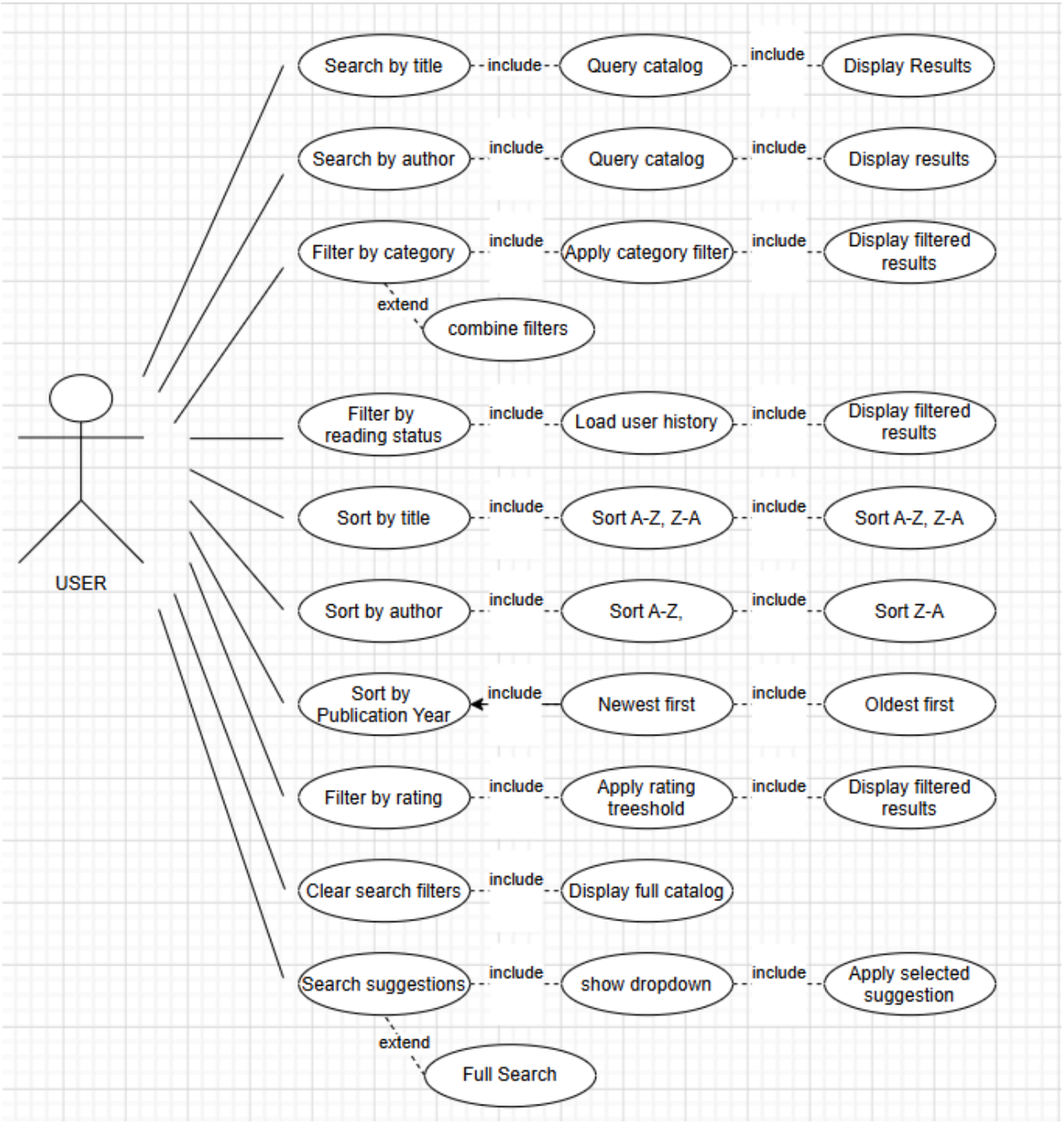
UC_10: Search suggestions

UC Name	UC_10 — Search suggestions
Dependency	None
Actors	Primary actor: User Secondary actor: None
Preconditions	The catalog contains at least one book.

Description of the Main Sequence	1. User starts typing in the search bar. 2. System displays a real-time dropdown of matching titles and authors. 3. User selects a suggestion. 4. System filters the catalog based on the selected suggestion.
Description of the Alternative Sequence	• No matches: Dropdown is not shown. • User ignores suggestions and presses Enter: Full search is performed.
Non functional requirements	Suggestions must appear within 300ms of typing. Minimum 2 characters required to trigger suggestions.
Postconditions	User has quickly located a book using an auto-suggested search term.

Use Case Diagrams

Search Filter



Functional Requirements

Req#	Role	Requirement	Comments	Priority	Date	Reviewed
FR_01	User	The system shall allow users to search books by title using a case-insensitive keyword match.	Partial matches must be supported. Results appear within 1 second.	1	01/04/2026	Migert Blloshm i
FR_02	User	The system shall allow users to search books by author name using a case- insensitive keyword match.	Partial name matches must be supported.	1	01/04/2026	Migert Blloshmi
FR_03	User	The system shall allow users to filter the catalog by one or more categories simultaneously.	Filter must apply within 0.5 seconds.	1	01/04/2026	Migert Blloshmi
FR_04	User	The system shall allow logged-in users to filter books by their personal reading status (Currently Borrowing, Returned, Reserved).	Status data must reflect real-time loan and reservation records.	2	01/04/2026	Migert Blloshmi
FR_05	User	The system shall allow users to sort the catalog by title in ascending or descending alphabetical order.	Sort must apply instantly (< 0.5s).	2	01/04/2026	Migert Blloshmi
FR_06	User	The system shall allow users to sort the catalog by author name in ascending or descending order.	Books without an author are displayed at the end.	2	01/04/2026	Migert Blloshmi
FR_07	User	The system shall allow users to sort the catalog by publication year (newest first or oldest first).	Books without a publication year are displayed at the end.	2	01/04/2026	Migert Blloshmi
FR_08	User	The system shall allow users to filter the catalog by minimum average rating.	Filter must apply within 0.5 seconds.	2	01/04/2026	Migert Blloshmi

FR_09	User	The system shall provide a 'Clear Filters' button that resets all active search terms and filters instantly.	Button must be disabled or hidden when no filters are active.	1	01/04/2026	Migert Blloshmi
FR_10	User	The system shall display real-time search suggestions in a dropdown when the user types at least 2 characters.	Suggestions must appear within 300ms. Dropdown closes on selection or outside click.	2	01/04/2026	Migert Blloshmi

Non-Functional Requirements

Req#	Category	Requirement
NFR_01	Performance	All search queries must return results within 1 second for a catalog of up to 10,000 books.
NFR_02	Performance	All filter and sort operations must apply within 0.5 seconds without a full page reload.
NFR_03	Performance	Search suggestions must appear within 300ms of the user typing at least 2 characters.
NFR_04	Usability	Search, filter, and sort controls must be clearly visible and accessible without scrolling on desktop and mobile.
NFR_05	Usability	Active filters must be visually indicated (e.g. highlighted badge or tag) so users always know what is applied.
NFR_06	Usability	The 'Clear Filters' button must be prominently displayed whenever any filter or search is active.
NFR_07	Reliability	Search and filter results must always reflect the current state of the catalog. Stale or cached results are not acceptable.
NFR_08	Scalability	The search and filter system must remain performant with up to 10,000 books and 200 concurrent users.
NFR_09	Accessibility	Search bar and filter controls must be keyboard-navigable and compatible with screen readers (WCAG 2.1 AA).
NFR_10	Security	Search input must be sanitized server-side to prevent SQL injection and XSS attacks.